Formula sheet for Quiz 9

Shapes and padding

A batch of ids is (N, T). `nn.Embedding(V, d)` maps it to (N, T, d) and holds $V \times d$ parameters. Reserved ids: `<pad>` $= 0$, `<unk>` $= 1$.
For sequences of lengths $\ell_1 \dots \ell_N$:

$$T = \max_i \ell_i, \qquad \text{cells} = N \times T, \qquad \text{padding share} = 1 - \frac{\sum_i \ell_i}{N \times T}$$

Masking

Mask is `ids` $\neq$ `PAD`. Masked mean:

$$\text{pooled}_i = \frac{\sum_t m_{i,t} E_{i,t}}{\sum_t m_{i,t}} \qquad \text{unmasked} \quad \frac{1}{T}\sum_t E_{i,t} \qquad \text{error} = 1 - \frac{\ell}{T}$$

With the pad vector at zero, the unmasked result is the masked one scaled by ℓ/T, which is where that error comes from.
For a per-token loss, `F.cross_entropy(..., ignore_index=0)`. Inside attention, a masked position is set to $-\infty$ before the softmax.

Convolution over a sequence

`nn.Conv1d` takes (N, C, T). A kernel of width k widens the receptive field by $k-1$ per layer, so L layers see
$L\,(k-1)+1$ positions, so $k = 3$ gives $2L + 1$.

Recurrence

$$h_t = \tanh(W_x x_t + W_h h_{t-1} + b)\,, \qquad h_0 = \mathbf{0}$$

LSTM. x_t is the input, h_t the hidden state, c_t the cell state. f_t is the forget gate, i_t the input gate, o_t the output gate, and $\tilde{c}_t$ the candidate.

$$\begin{aligned} f_t &= \sigma(W_f[h_{t-1}, x_t] + b_f) \\ i_t &= \sigma(W_i[h_{t-1}, x_t] + b_i) \\ o_t &= \sigma(W_o[h_{t-1}, x_t] + b_o) \\ \tilde{c}_t &= \tanh(W_c[h_{t-1}, x_t] + b_c) \end{aligned}$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, \qquad h_t = o_t \odot \tanh(c_t)$$

Backpropagation through time:

$$\frac{\partial h_T}{\partial h_t} = \prod_{k=t+1}^{T} \frac{\partial h_k}{\partial h_{k-1}}$$

a product of $T - t$ terms, so a per-step factor r gives r^n after n steps. A recurrent layer takes T sequential steps with total work growing as T; self-attention takes 1 sequential step with work growing as T^2.

Attention

$\mathbf{Q} = \mathbf{X}\mathbf{W}_\mathbf{Q}$, $\mathbf{K} = \mathbf{X}\mathbf{W}_\mathbf{K}$, $\mathbf{V} = \mathbf{X}\mathbf{W}_\mathbf{V}$.

$$\mathbf{A}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = S\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V}$$

The weight matrix is $n \times n$ and each row sums to 1. Multi-head with h heads of width d/h:
$\text{MultiHead}(\mathbf{X}) = [\text{head}_1; \dots; \text{head}_h]\mathbf{W}_\mathbf{O}$.

Positional encoding

$$P(k, 2i) = \sin(k/n^{2i/d}), \qquad P(k, 2i+1) = \cos(k/n^{2i/d})$$

k is the position, i the dimension pair, $n = 10{,}000$. The learned alternative is a trainable table with one row per position.

The block

$$\mathbf{x} \leftarrow \text{Norm}(\mathbf{x} + \text{Attention}(\mathbf{x})), \qquad \mathbf{x} \leftarrow \text{Norm}(\mathbf{x} + \text{FFN}(\mathbf{x}))$$

The feed forward network widens to $4d$ and back. BatchNorm takes μ, σ over the **batch** axis, LayerNorm over the **feature** axis; both then scale and shift by learned parameters.